PRD: Trending Products E-commerce Platform

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1. Overview

We are building an online store where the client lists trending, fast-selling products, and customers browse full product details and pay online (UPI, cards, netbanking, wallets) or by Cash on Delivery (COD). The client runs everything from an admin dashboard: products with GST details, categories, stock, offers, orders and reports.

Problem. The client sells fast-moving products and needs to launch listings quickly, push prepaid orders (cheaper to fulfil, no COD returns), and raise order value with bundle-style discounts, all with GST-compliant billing.

What makes this product different from a generic store:

- A prepaid-first checkout: customers see a clear discount for paying online.
- A cart-value and quantity-based discount engine for multi-product purchases.
- GST captured at product level and printed correctly on every invoice.
- A fast product-entry flow, since the catalogue changes often with trends.

Document status. Draft v0.1. Assumptions are marked and collected in Section 15 for the client to confirm.

2. Goals and success metrics

The store succeeds if the client can list a product in under 5 minutes, and customers can find, trust and pay for it with few steps.

Business goals

- Launch a fully working store with online payment and COD.
- Raise the share of prepaid orders through a visible prepaid discount.
- Raise average order value (AOV) through multi-product discounts.
- Issue GST-compliant invoices automatically for every order.

Success metrics (targets are proposals; confirm with the client)

Metric	Target	How measured
Prepaid share of orders	60% or more within 3 months	Orders by payment method
Average order value	15% higher than single-item baseline	Dashboard report
Checkout conversion	2.5% or more of sessions	Analytics funnel
Product page load (LCP, 4G mobile)	Under 2.5 seconds	Lighthouse / RUM
Time to add one product	Under 5 minutes	Admin usability test
COD return-to-origin (RTO) rate	Under 20%	Shipment reports

Non-goals for v1: multi-vendor marketplace, mobile apps, multi-language, international shipping, loyalty points.

3. Users and personas

There are two main groups: shoppers who buy, and the client's team who run the store.

Persona	Who they are	What they need
Shopper (guest or registered)	Mobile-first buyer in India, price-sensitive, impulse-driven	Fast pages, clear photos and details, trust signals, easy UPI and COD checkout, order tracking
Owner / Admin	The client	Full control: products, offers, orders, staff, reports, settings
Catalogue manager	Staff who add and edit products	Quick product entry, bulk upload, stock updates
Order / Support executive	Staff who process orders	Order list, status updates, invoices, refunds, customer details

Admin roles and permissions. Roles are enforced on the server. The Owner can create staff and choose what each role can see or change (for example, a Catalogue manager cannot issue refunds or view revenue reports).

4. Scope

Version 1 delivers one storefront, one admin dashboard and one API for a single seller in India, selling in INR.

In scope for v1

- Customer storefront (responsive web, mobile-first)
- Admin dashboard with role-based access
- Product, category, brand and inventory management with GST fields
- Offers: prepaid discount, multi-product discounts, coupons
- Checkout with online payment and COD
- Order management, shipping integration, returns and refunds
- GST tax invoices and GST reports
- Email, SMS and WhatsApp order notifications (transactional)
- Basic sales and product reports

Out of scope for v1

- Multi-vendor marketplace
- Native mobile apps
- Multiple currencies and languages
- International shipping and taxes
- Loyalty points, subscriptions, reviews moderation by AI

Assumptions

- The client is GST-registered and sells within India.
- A single warehouse ships all orders.
- Razorpay (or a similar gateway) and Shiprocket (or a similar aggregator) are acceptable.
- The client supplies product photos and descriptions.

5. Storefront requirements

The storefront must let a shopper go from landing to paid order in as few taps as possible, on a phone.

ID	Feature	Requirement
S-1	Home page	Banners, Trending Now, Best Sellers, New Arrivals, category tiles, active offers strip. All sections controlled from admin.
S-2	Category and listing pages	Nested categories, grid with price, discount badge, rating, stock tag. Sort (popularity, price, newest) and filters (price, brand, category attributes). Pagination or infinite scroll.
S-3	Search	Search bar with suggestions, typo tolerance, and search by name, brand or SKU.
S-4	Product detail page	Image gallery with zoom and video, title, price with MRP and discount percent, tax-inclusive note, variants (size, colour), stock status, full description, specification table, highlights, delivery pincode check with estimated date, return policy, ratings and reviews, related products, share button.
S-5	Offer visibility	Product page shows applicable offers, for example "Extra 5% off on prepaid" and "Buy 2 save 10%".
S-6	Cart	Edit quantity, remove, save for later, live price breakdown with offer savings shown line by line, coupon field, tax and shipping preview.
S-7	Checkout	Guest or OTP login, address form with pincode auto-fill, shipping method, payment choice (online or COD), order summary. See Section 8.
S-8	Customer account	Login by mobile OTP or email, profile, saved addresses, order history, invoice download, track order, cancel and return requests, wishlist.
S-9	Content pages	About, contact, FAQ, shipping, returns, privacy, terms. Editable from admin.
S-10	SEO and sharing	Clean URLs, meta tags, structured data (Product, Offer), sitemap, social preview images.

Trust and conversion features: secure-payment badges, COD availability indicator by pincode, low-stock messages, and recently viewed products.

6. Admin dashboard and product management

The admin must be able to add a complete, GST-ready product on a single screen and publish it immediately.

Dashboard modules: Home (today's sales, orders, low stock), Products, Categories, Inventory, Orders, Customers, Offers, Content, Reports, Staff and roles, Settings, Audit log.

Add-product form fields

Group	Fields
Basic	Name, slug (auto), brand, category (one or more), tags, status (draft, active, archived)
Trending flags	Mark as Trending, Best Seller, New Arrival, Featured; sort order on home page
Content	Short description, full description (rich text), highlights, specifications (key and value rows), video link
Media	Multiple images with drag to reorder, one cover image, optional per-variant image
Pricing	MRP, selling price, cost price (admin only), auto-calculated discount percent
GST and tax	HSN code, GST rate (0, 5, 12, 18 or 28 percent), price includes GST (yes or no), GST-exempt toggle, cess if applicable
Variants	Options (size, colour), one row per combination with SKU, price, stock, barcode, image
Inventory	Stock quantity, low-stock alert level, allow backorder (yes or no)
Shipping	Weight, length, width, height, COD allowed (yes or no), returnable (yes or no), return window in days
Offers	Eligible for prepaid discount, eligible for multi-buy discount, attach a product-level offer
SEO	Page title, meta description, URL slug

Rules

- HSN code and GST rate are mandatory before a product can be set to active.
- The system computes base price and GST from the selling price. Example: at an inclusive price of Rs 1,180 and 18 percent GST, base price is Rs 1,000 and GST is Rs 180.
- Bulk upload by CSV or Excel template (create and update), with row-level error report.
- Duplicate product, bulk edit price and stock, bulk status change.
- Category management: nested tree, image, SEO fields, per-category attribute sets (for example Clothing has size and fabric).
- Every change is written to the audit log with who, what, before and after.

7. Offers and discount engine

The admin creates offers from rules, with no code changes, and the cart applies them automatically and shows the saving.

Offer types required for v1

Offer	Example	Configurable settings
Prepaid discount	5% off (max Rs 200) when paying online	Percent or flat amount, maximum cap, minimum cart value, applies to all, categories or chosen products
Multi-product quantity discount	Buy 2 get 10% off, buy 3 get 15% off	Quantity tiers, same product or any products, category or storewide
Cart-value discount	Rs 150 off above Rs 1,500	Spend tiers, percent or flat
Bundle or combo	Product A plus Product B for Rs 999	Fixed bundle price, chosen products
Buy X get Y	Buy 2 get 1 free	Cheapest item free, or a specified item
Coupon code	WELCOME10	Code, percent or flat, usage limit total and per customer, first order only, start and end date
Product-level sale	Flash sale price on one product	Sale price, start and end time, countdown on product page

Rules for applying offers

1. Every offer has a start and end date and an active toggle, and can be limited to chosen products, categories or customer groups.

2. Each offer has a priority and a stackable flag. By default, one item-level offer applies per item, and the prepaid discount stacks on top of it.
3. A maximum total discount per order can be set to protect margin.
4. The prepaid discount is applied only when online payment is selected. Switching to COD at checkout removes it, and the cart shows the amount the customer would save by paying online.
5. Discounts are never applied below cost price unless the admin allows it.
6. GST is calculated on the discounted price, and the discount is spread across items proportionally.
7. If a prepaid order is refunded, the discount is refunded proportionally. If a prepaid order fails and the customer retries with COD, the discount is not kept.

Worked example. Cart has 3 items at Rs 500 each (Rs 1,500). Buy 3 get 15% off gives Rs 225 off, leaving Rs 1,275. Paying online adds 5% prepaid discount on Rs 1,275, which is Rs 63.75 off, for a final Rs 1,211.25 before shipping.

8. Checkout and payments

Checkout offers two payment paths, online and COD, and nudges the shopper toward online with a visible saving.

Checkout steps

1. Login by OTP or continue as guest (mobile number required).
2. Enter or pick a delivery address. The pincode is checked for serviceability and COD availability.
3. Choose a shipping method, if more than one is offered.
4. Choose payment: Pay Online (with the prepaid saving shown) or Cash on Delivery.
5. Review the price breakdown and place the order.

Online payment

- Gateway: Razorpay (proposed). Supports UPI, cards, netbanking and wallets.
- Order is created as Pending Payment, and is confirmed only after a server-verified payment webhook, never from the browser redirect alone.
- Payment signature verification, idempotent webhook handling, and automatic retry if the payment fails.
- Unpaid online orders expire after a set time (for example 30 minutes) and release reserved stock.
- Refunds go back to the original payment method from the admin, full or partial.

Cash on Delivery

- Admin can switch COD on or off globally, per product, per pincode, and by order value range (minimum and maximum).
- Optional COD handling fee (flat amount), shown clearly in the price breakdown.
- Fraud control: mobile OTP verification before a COD order is confirmed, a block list for customers with repeated refused deliveries, and an optional rule to require prepaid above a set order value.
- Order is confirmed immediately, and payment is marked Pending until the courier remits the cash.

Price breakdown shown to the shopper: item total (MRP), product discounts, multi-product discount, coupon, prepaid discount, subtotal, GST (included or shown separately), shipping charge, COD fee, and final payable amount.

Shipping charges: free above a threshold, flat rate, or weight-based, configurable in admin settings.

9. Order management and fulfilment

Staff can see every order, move it through clear statuses, and ship it without leaving the dashboard.

Order statuses

Status	Meaning	Applies to
Pending Payment	Waiting for online payment	Prepaid
Confirmed	Payment verified, or COD order verified	Both
Processing	Being packed	Both
Shipped	Handed to courier, tracking number added	Both
Out for Delivery	Courier update	Both
Delivered	Received by customer; COD cash marked collected	Both
Cancelled	Cancelled by customer or admin before shipping	Both
Return Requested, Returned	Customer return flow	Both
RTO	Return to origin, delivery failed or COD refused	Mostly COD

Status	Meaning	Applies to
Refunded	Money returned (full or partial)	Both

Admin order features

- Order list with filters (status, payment method, date, pincode, customer) and search by order number, phone or email.
- Order detail: items, price breakdown, addresses, payment record, timeline, internal notes.
- Actions: confirm, cancel, edit address before shipping, add tracking, print invoice and packing slip, partial or full refund, mark COD collected.
- Bulk actions: print invoices, generate shipping labels, change status.

Shipping integration. Integrate one aggregator (Shiprocket proposed) for rates, pickup, label, tracking and webhooks. Shoppers get tracking updates by SMS, WhatsApp or email.

Returns and refunds. Shopper requests a return within the product's window with a reason and photos. Admin approves, courier picks up, stock is restocked on inspection, and the refund is issued (to source for prepaid, to bank or UPI for COD).

COD remittance. Track courier COD payouts against delivered COD orders so the client can reconcile cash received.

Stock handling. Stock is reserved when an order is placed, deducted on shipment, and returned on cancellation or return. Every change is logged in a stock ledger.

10. GST and invoicing

Every order gets a correct GST tax invoice, with tax split chosen by the delivery state.

Store settings (one time): legal business name, GSTIN, registered address, state, PAN, invoice prefix and numbering series, logo, authorised signature, terms.

Tax calculation rules

Case	Tax applied
Delivery state equals seller state	CGST plus SGST, each half of the GST rate
Delivery state differs from seller state	IGST at the full GST rate
Product marked GST-exempt (0 percent)	No tax, shown as exempt on the invoice

- Tax is computed per line item on the price after all discounts, using the item's HSN code and GST rate.

- Prices can be stored tax-inclusive or tax-exclusive, and the storefront shows tax-inclusive prices.
- Shipping and COD fees are taxed at the applicable rate and shown as separate lines.
- Rounding follows one consistent rule, and the invoice shows the round-off line.

Tax invoice contents: seller name, address and GSTIN; invoice number and date; buyer name, address and state; order number; item description, HSN, quantity, unit price, discount, taxable value, CGST, SGST or IGST rate and amount; total in numbers and words; payment mode; place of supply; authorised signatory.

Documents generated automatically: tax invoice on dispatch, credit note on return or refund, and a packing slip. Invoices are PDF, emailed to the customer and downloadable from the account page and admin. Invoice numbers are sequential and never reused.

GST reports in admin: sales register by date, tax summary by rate (HSN-wise summary), state-wise tax split, and CSV export for the accountant to prepare GSTR-1 and GSTR-3B. Filing itself is out of scope.

Open point: if the client wants e-invoicing (IRN) or e-way bills, this needs a GST Suvidha Provider integration and is an add-on. See Section 15.

11. Data model

The database is PostgreSQL, with money stored as integer paise and order lines storing snapshots of product data.

Entity	Key fields
Product	name, slug, brand, status, description, specs, trending flags, hsn_code, gst_rate, price_includes_gst, cod_allowed, returnable, return_days, weight, dimensions, SEO
Variant	product, sku, mrp, selling_price, cost_price, barcode, option values, active
Category	parent, name, slug, image, sort order, attribute set
Image	product, variant (optional), url, sort order
Inventory level and movement	variant, quantity, reserved, low-stock level; ledger rows with change, reason, reference
Customer and Address	name, mobile, email, blocked flag; address lines, pincode, state

Entity	Key fields
Cart and Cart item	customer or session, variant, quantity, applied offers
Offer	type, rule config (JSON), scope, priority, stackable, usage limits, start, end, active
Coupon	code, offer, per-customer limit, used count
Order	number, customer, status, payment method, payment status, subtotal, discount breakdown, tax breakdown, shipping, COD fee, total, addresses
Order item	variant, name, SKU, HSN, GST rate, unit price, quantity, discount, taxable value, CGST, SGST, IGST
Payment and Refund	provider, reference, amount, status, method; refund amount, reason, status
Shipment	courier, AWB, status, tracking events, COD amount, remittance status
Invoice and Credit note	number, order, PDF url, date, totals
Staff, Role, Permission	login, role, permission codes such as products.write and orders.refund
Audit log	staff, action, entity, before and after values, time
Settings	store details, GSTIN, shipping rules, COD rules, payment keys

Key design rules: order items snapshot name, SKU, price, HSN and tax so past orders never change; soft delete for products, categories and customers; every stock change goes through the ledger.

12. Non-functional requirements

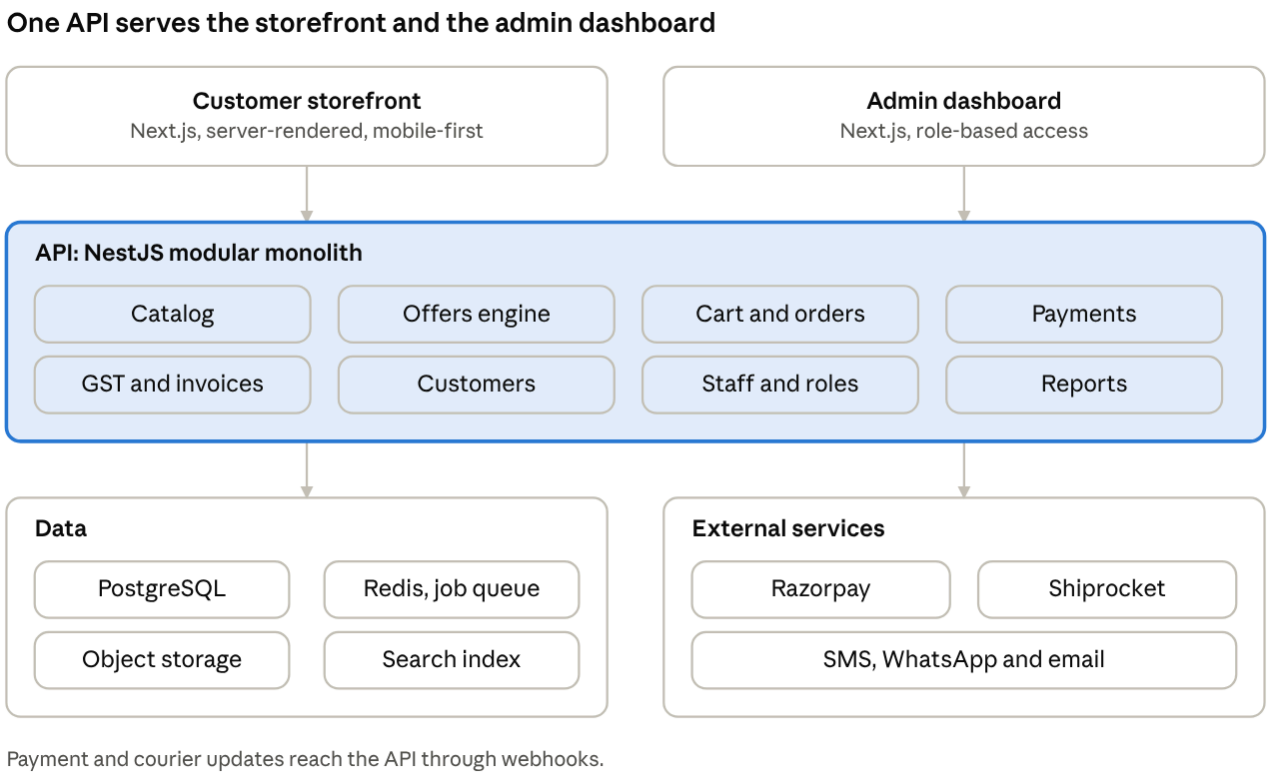
The store must be fast on mobile, safe with payments and personal data, and able to handle sale-day traffic spikes.

Area	Requirement
Performance	LCP under 2.5 s on 4G mobile; API responses under 300 ms at the 95th percentile for catalogue reads; images in WebP or AVIF with lazy loading and CDN

Area	Requirement
Scalability	Handle 100 concurrent shoppers at launch, and scale horizontally to 10x for flash sales without redesign; cache catalogue and home pages
Availability	99.5% monthly uptime; daily database backups with point-in-time recovery; tested restore
Security	HTTPS everywhere; hashed passwords; short-lived tokens; role checks on the server; input validation; rate limiting on login, OTP and checkout; protection against SQL injection, XSS and CSRF
Payments	Card data never touches our servers (gateway-hosted); webhook signature checks; idempotent payment handling
Privacy	Compliance with India's Digital Personal Data Protection Act: consent, privacy policy, data export and deletion on request
SEO	Server-rendered product and category pages, structured data, sitemap, canonical URLs
Accessibility	WCAG 2.1 AA for key flows
Browser support	Last 2 versions of Chrome, Safari, Edge, Firefox; Android and iOS browsers
Observability	Error tracking, logs, uptime alerts, payment and webhook failure alerts
Admin safety	Audit log, confirmation on destructive actions, soft delete
Maintainability	TypeScript, automated tests on pricing, offers and GST logic, CI pipeline, staging environment

13. Technical architecture and stack

The system is a modular monolith: one API with a module per business area, used by both the storefront and the admin dashboard.



system architecture · 3 layers

Both front ends call the same API; data stores and third-party services sit behind it.

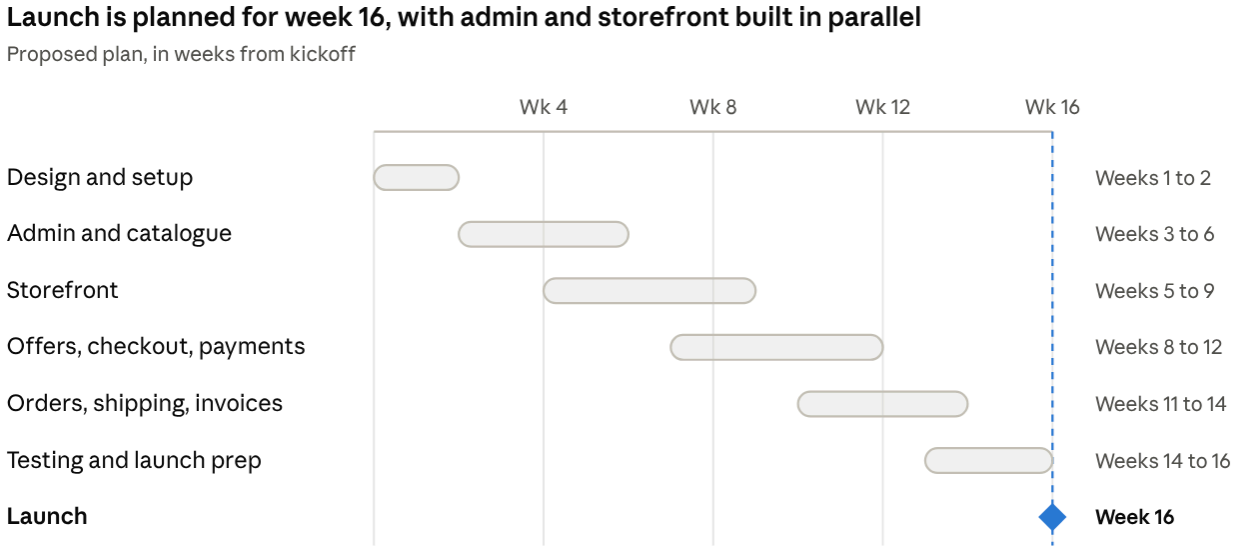
Layer	Choice	Reason
Storefront	Next.js, TypeScript, Tailwind	Server-rendered pages for SEO and fast mobile load
Admin dashboard	Next.js, shadcn/ui, TanStack Table, React Hook Form, Zod	Fast build of data-heavy screens and forms
API	NestJS (REST), JWT access and refresh tokens	Clear modules, shared TypeScript types
Database	PostgreSQL with Prisma	Reliable transactions for orders, stock and tax
Cache and jobs	Redis with BullMQ	Caching, CSV imports, emails, invoice PDFs, webhook retries
Search	Meilisearch	Typo-tolerant search with filters
File storage	S3-compatible storage with CDN	Product images and invoice PDFs, presigned uploads
Payments	Razorpay	UPI, cards, netbanking, wallets, refunds

Layer	Choice	Reason
Shipping	Shiprocket	Rates, labels, tracking, COD remittance data
Notifications	Email, SMS and WhatsApp providers	Order and shipping updates
Hosting and delivery	Docker on a cloud host in India (Mumbai region), GitHub Actions CI, staging and production	Low latency for Indian shoppers, repeatable releases
Monitoring	Sentry, uptime checks, log aggregation	Fast detection of payment and checkout errors

Code layout: one monorepo with apps for the storefront, admin and API, and a shared package holding validation schemas, types and permission codes.

14. Release plan

We propose a 16-week build, with the admin and storefront developed in parallel after the first six weeks.



release plan · 6 phases, launch in week 16

Phases overlap from week 5, and the last two weeks are reserved for testing, client acceptance and go-live.

Phase	Delivers	Done when
Design and setup	Final scope sign-off, UI designs for storefront and admin, repo, CI, staging, database, auth and roles	Client approves designs and the Section 15 questions are answered
Admin and catalogue	Categories, attributes, products with GST fields, variants, images, inventory, CSV import, audit log	Client can add a full GST-ready product in under 5 minutes
Storefront	Home, listing, search, product page, cart, customer accounts	Shoppers can browse and fill a cart on mobile
Offers, checkout, payments	Offer engine, prepaid and multi-product discounts, coupons, checkout, Razorpay, COD rules	Test orders pass for online and COD with correct discounts and tax
Orders, shipping, invoices	Order management, Shiprocket, returns and refunds, GST invoices, credit notes, reports, notifications	An order moves from placement to delivery, with invoice, end to end
Testing and launch prep	Automated and manual testing, load test, security review, client acceptance, data load, go-live	Acceptance criteria in Section 15 are met

Planning assumptions: a team of two full-stack developers and one designer, with weekly demos to the client. Timelines are estimates until scope and open questions are confirmed, and delays in content, approvals or gateway and courier onboarding move the launch.

15. Risks, open questions and acceptance criteria

The plan holds if the client answers the open questions in the first two weeks and the main risks below are managed.

Open questions for the client

- [ ] What product categories and approximate catalogue size at launch (10, 100 or 1,000 or more products)?
- [ ] Is the business GST-registered, and will it sell only within India? Is e-invoicing (IRN) or e-way bill needed?
- [ ] Should prices be shown tax-inclusive (assumed) or tax-exclusive?
- [ ] What exact prepaid discount is wanted (percent or flat, any cap)?

- [ ] What multi-product discount tiers are wanted (for example buy 2 save 10%, buy 3 save 15%)? Should they combine with coupons?
- [ ] COD rules: any maximum order value, COD fee, and pincodes where COD is blocked?
- [ ] Which payment gateway and courier accounts exist already, or should we use Razorpay and Shiprocket?
- [ ] Does the client need guest checkout, login by OTP only, or both?
- [ ] Who supplies product photos, descriptions and brand assets, and when?
- [ ] Is a domain, hosting account and brand identity ready? Who owns hosting costs?
- [ ] Is a return window needed for all products, or does it vary?
- [ ] Will staff beyond the owner use the admin, and with what roles?

Risks

Risk	Impact	Mitigation
Scope grows beyond this document	Launch slips	Phase-wise sign-off, written change requests, v2 backlog
Offer rules conflict or stack unexpectedly	Margin loss	Priority and stack flags, per-order discount cap, automated tests with worked examples
High COD returns (RTO)	Shipping loss	OTP check on COD, pincode and value limits, prepaid discount, block list
Wrong GST or HSN data	Compliance issues	Mandatory fields, accountant review of invoice sample, tax unit tests
Payment webhook failures	Paid orders not confirmed	Idempotent webhooks, reconciliation job, alerts
Gateway or courier onboarding delays (KYC)	Launch blocked	Start account applications in week 1
Flash-sale traffic spikes	Slow or failing site	Caching, CDN, load test before launch
Late content from the client	Empty store at launch	Bulk upload template and content deadline by week 6

Acceptance criteria for v1

1. An admin can add a product with HSN, GST rate, variants, images and stock, and it appears on the storefront immediately.
2. A shopper can find a product, view full details, and place an order by online payment and by COD on a phone.
3. The prepaid discount applies only to online payment and is removed if COD is chosen.
4. A cart with several products receives the configured multi-product discount, with the saving shown line by line.
5. Each delivered order has a GST invoice with correct CGST, SGST or IGST, and refunds produce a credit note.
6. Staff roles block access to modules and actions they are not permitted to use.
7. The GST sales report matches the sum of invoices for a test period.
8. Product pages meet the performance and security targets in Section 12.